

Fig. 3: Compatibility requirement

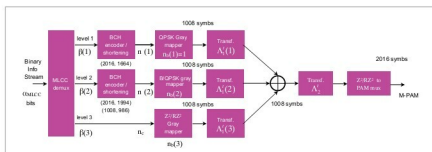


Fig. 4: PHY technology for Gigabit Ethernet over POF

MAC interface (SGMII/RGMII/GMII), which is defined through IEEE standardisation process. Gigabit Ethernet is targeting to penetrate the vehicle market in 2018.

Two essential factors are completion of IEEE standards and a roadmap for the future. With the completion of IEEE 100BASE-T standard, chip vendors expect to follow the trends in automotive Ethernet. Upcoming 100BASE-T1 standard will be based on BradR-Reach specifications. Gigabit automotive Ethernet PHY transceiver chip is based on 1000BASE-T1 draft standard.

Advantages of 1000BASE-T1 are:

- Supports machine vision for ADAS and is suitable for high-speed video links
- Supports more video channels and can uplink 100Mbps domains
- Can use cost-effective unshielded cables

- Ethernet cables are lightweight and flexible
- Possibility to support 100BASE-T1 and 1000BASE-T1, and can define systems with choice of resolution by replacing a single module
- Adds video paths through software

Uncompressed high-definition video needs greater than 100Mbps. At 100Mbps, strong compression is required that can degrade images. Compression algorithms add latency, which limits the applications for the Ethernet. At 1Gbps, little or no compression is required. Low latency supports higher quality image transport. For machine vision, high-definition navigation and instrument cluster that has a high bandwidth compression algorithm are required.

The standard that defines 100BASE-T1 was developed outside of IEEE 802.3 committee's governance. But it is strictly based on

IEEE 802.3 framework. PHY is fully compatible with a standard Ethernet MAC that allows seamless interoperability with nearly any Ethernet-capable microcontroller or embedded system element, provided its MAC's Media Independent Interface (MII) is compatible with One-Pair Ethernet PHY.

Maintaining compatibility also eliminates the need for any special drivers, making it easily interoperable with Ethernet devices that support IEEE 1588 remote timing and clock synchronisation protocol. Precise timing and synchronisation that IEEE 1588 provides will be critical in automotive sub-systems such as braking, engine control, active suspension and anywhere else where real-time operations are performed.

Incoming data is divided into blocks that are partitioned in up to three levels with different degrees of coding and then mapped to a 2D constellation. Bits in the first level are protected with a BCH code that provides powerful error correction, while the second level is protected with another BCH code and the last level is not coded. These are then mapped such that those from levels two and three are far apart (that is, coset partitioning).

The first level is always enabled by using QPSK modulation. The second level, depending on bit-rate configuration, uses BPSK or QPSK modulations. For BPSK, BCH code is shortened to 1008-bit length. The third level directly maps the information into configurable size modulations over Z2 or RZ2 lattices. After mapping, 2D lattice transformations are used to implement the coset partitioning. Finally, MPAM symbols are generated by time multiplexing from 2D symbols.

This is a process for configuration in which the numbers of bits in each level are 1, 1 and 1.5. This corresponds to 16-PAM modulation and bit rate of 1Gbps, with symbol